PROJECT SYNOPSIS

Project Title

<u>Cyber Sense AI</u>: An AI-Powered Human Cyber Behaviour Prediction and Personalized Cybersecurity Coaching System

1. Introduction

With the rapid growth of digital communication and online services, cyber threats such as phishing, malware, malicious URLs, credential theft, and social engineering attacks have become increasingly common. Reports consistently show that human behaviour is one of the primary causes of successful cyberattacks. Most existing cybersecurity solutions are reactive, meaning they detect or respond to threats only after they occur. Moreover, these solutions often focus on a single type of threat and provide limited guidance to users for improving their cybersecurity practices.

Cyber Sense AI is proposed as an AI-powered cybersecurity platform that proactively predicts human cyber behaviour risks before cyber incidents occur. The system analyses different aspects of user cyber behaviour, including SMS phishing, email phishing, malicious URL interaction, password security, authentication behaviour, and behavioural changes over time. Using Artificial Intelligence, Machine Learning, and Explainable AI (XAI), the platform not only predicts cyber risks but also explains the reasons behind every prediction and provides personalized cybersecurity coaching to help users improve their cyber hygiene.

The project follows a modular architecture where each cyber risk is analysed independently, and the individual risk scores are combined to generate an overall Cyber Risk Score representing the user's cybersecurity posture.

2. Problem Statement

Existing cybersecurity solutions primarily focus on detecting malware, spam emails, malicious websites, or network intrusions after they have already occurred. They rarely consider human cyber behaviour as a major factor contributing to cyber incidents. Additionally, most available systems are designed to solve only one specific cybersecurity problem and do not provide a comprehensive assessment of a user's cyber risk.

The major limitations of existing systems include:

- Reactive detection instead of proactive prediction.
- Limited focus on human behavioural factors.
- Lack of Explainable AI for understanding predictions.
- Generic cybersecurity awareness programs.
- No personalized cybersecurity recommendations.
- Absence of an integrated cyber risk assessment platform.

Therefore, there is a need for an intelligent system capable of predicting multiple human cyber behaviour risks, explaining AI decisions, and providing personalized guidance to reduce future cyber incidents.

3. Objectives

The primary objectives of Cyber Sense AI are:

- To develop an AI-powered platform for predicting human cyber behaviour risks.
- To detect phishing SMS using machine learning techniques.
- To identify phishing emails and malicious URLs.
- To assess password security and authentication behaviour.
- To detect behavioural drift by analysing changes in user activities over time.
- To calculate an overall Cyber Risk Score by combining multiple risk factors.
- To integrate Explainable AI (SHAP/LIME) for transparent and interpretable predictions.
- To provide personalized cybersecurity coaching based on individual user behaviour.

- To improve cybersecurity awareness and encourage safer online practices.

4. Scope of the Project

The proposed system covers the following functional areas:

- SMS Phishing Detection
- Email Phishing Detection
- Malicious URL Detection
- Authentication Behaviour Analysis
- Behavioural Drift Detection
- Explainable AI for prediction transparency
- Personalized Cybersecurity Coaching
- Cyber Risk Dashboard
- Historical Risk Reports

The project is designed with a modular architecture, making it easy to integrate additional cybersecurity risk modules in the future.

5. Existing System

Current cybersecurity tools generally include spam filters, antivirus software, password strength checkers, URL reputation services, and intrusion detection systems. While these tools are effective in their individual domains, they function independently and often lack an integrated approach to analysing human cyber behaviour.

Limitations of Existing Systems

- Focus only on individual cyber threats.
- Reactive rather than proactive.
- Limited behavioural analysis.
- No overall cyber risk assessment.

- Lack of Explainable AI.
- Generic awareness programs without personalization.

6. Proposed System

Cyber Sense AI proposes an intelligent website-based cybersecurity platform that predicts cyber risks before incidents occur by analysing user behaviour across multiple cybersecurity domains.

The platform consists of specialized AI modules that independently assess different cyber risks such as SMS phishing, email phishing, malicious URLs, password security, authentication behaviour, and behavioural drift. The outputs of these modules are combined through a Risk Fusion Engine to calculate an Overall Cyber Risk Score.

The system incorporates Explainable AI to provide transparent reasons for every prediction and delivers personalized cybersecurity coaching based on the user's risk profile, enabling users to improve their cybersecurity habits proactively.

7. Modules of the System

The proposed system consists of the following modules:

- User Registration and Authentication
- SMS Phishing Detection
- Email Phishing Detection
- Malicious URL Detection
- Authentication Behaviour Analysis
- Behavioural Drift Detection
- Explainable AI Module
- Overall Cyber Risk Dashboard
- Personalized Cybersecurity Coaching
- Reports and Risk History

8. Technologies Used

Frontend

- HTML
- CSS (Bootstrap)
- React (TypeScript)

Backend

- Flask (Python)

Machine Learning

- Scikit-learn
- TensorFlow
- LSTM
- Random Forest
- XGBoost

Explainable AI

- SHAP
- LIME

Database

- MySQL

Development Tools

- Visual Studio Code
- GitHub
- Figma

9. Datasets

The project utilizes publicly available datasets from Kaggle, including:

- SMS Spam Collection Dataset

- Email Phishing Dataset
- Phishing URL Dataset
- Password Strength Dataset
- CERT Insider Threat Dataset
- Cybersecurity Awareness Dataset

These datasets will be modified and extended to suit the requirements of Cyber Sense AI and to support individual risk prediction modules.

10. Expected Outcomes

The proposed system is expected to:

- Predict cyber risks before attacks occur.
- Detect multiple categories of cybersecurity threats.
- Improve users' cybersecurity awareness.
- Provide transparent AI predictions through Explainable AI.
- Deliver personalized recommendations for reducing cyber risks.
- Generate an Overall Cyber Risk Score reflecting the user's cybersecurity posture.
- Promote safer online behaviour and better cyber hygiene.

11. Future Enhancements

Future versions of Cyber Sense AI may include:

- Browser extension for real-time phishing detection.
- Mobile application for Android and iOS.
- Integration with enterprise security systems.
- Real-time behavioural monitoring.
- Cloud deployment.
- Federated learning for privacy-preserving AI.

- Integration with threat intelligence platforms.
- Generative AI assistant for cybersecurity guidance.

12. Conclusion

Cyber Sense AI is designed as a proactive, AI-driven cybersecurity platform that addresses the growing challenge of human-centric cyber threats. By combining behavioural analytics, machine learning, Explainable AI, and personalized cybersecurity coaching, the system aims to move beyond traditional reactive security approaches and help users recognize, understand, and reduce their cyber risks before they lead to security incidents. Its modular architecture also enables future expansion, making it suitable for both academic research and practical cybersecurity applications.